

Three-Layer Quantum-Classical Information Hierarchy

Grounded in cryptochrome (CRY) radical pair mechanism

LAYER 1 : Nuclear Spin Quantum Memory

Molecule

FAD in cryptochrome
 ^{31}P on FMN-phosphate

$I = 1/2$, 100% abundance

862 Hz
3.6 peV



Coherence

Diamagnetic: $T_2 = 3.2$ ms

RP-coupled: $T_2 = 160$ μs

(r^{-6} PRE, $r_{\text{eff}} = 7$ $\text{\AA}$)

Quantum information:

^{31}P spin state encodes B -field direction $\rightarrow$ modulates S-T mixing rate via HFC

HFC: $A = 200$ MHz = 0.83 μeV

^{31}P nuclear spin $\leftrightarrow$ electron spin

LAYER 2 : Radical Pair Quantum-Classical Interface

Blue light (450 nm)

FAD $\rightarrow$ $[\text{FAD}^{\bullet-} \cdots \text{Trp}^{\bullet+}]$

via Trp triad/tetrad

$r = 15\text{-}20$ $\text{\AA}$, $|J| < 1$ MHz

$\tau_{\text{RP}} = 1\text{-}10$ μs

HFC-driven
mixing



Magnetic field effect

MFE = -5.6% @ 0.29 mT

MFE = -0.55% @ Earth field

$\Phi_S(B)$ = singlet yield

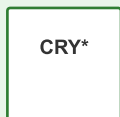
Quantum-to-classical transduction:

Nuclear spin state $\rightarrow$ S-T mixing rate $\rightarrow$ $\Phi_S(B)$ $\rightarrow$ product branching ratio

$\Delta\Phi_S = 0.55\%$ (Earth field)

No homeostatic buffering
(conformational, not metabolite)

LAYER 3 : Classical Circadian Signalling



Photoactivated

PER

Binding

CLOCK
/BMAL1

Repression

Gene
expr.

Oscillation

Period
shift

2.7-8.2 min

Circadian rhythm $\rightarrow$ mood, sleep, seasonal behaviour

Key: RP confirmed ($|J| < 1$ MHz) | No homeostatic buffering | Circadian amplification (limit-cycle)

CRY is the only currently defensible RPM host in the human brain

3.6 peV

14 orders of magnitude
0.83 μeV

-eV